Notes – Precedence in Expression Evaluation **Name _______________________**

For each expression below,

a) **Parenthesize if necessary,**
b) **Label each in operand and operator in order of evaluation**
c) **Build an expression tree**
d) **Evaluate expression or give final values of variables as indicated**

Note: Assume undefined methods and variables are appropriately defined.

Use int a=2, b=4, c=5, d=3;

`1. a + (b + c) * d`

Evaluation order:

a	+	(b	+	c)	*	d
1	7	2	4	3	6	5

Expression Tree:

```
                    +
        a                       *
                            +           d
                         b     c
```

Value: 29

Use int a=2, b=4, c=5, d=3;

`2. c + a * Math.round(2.6 + b)`

Use int a=2, b=4, c=5, d=3;

`3. (a + b) + ++b`

Use int a=2, b=4, c=5, d=3;

`4. c /= ++c;` ← This actually includes the type cast shown below:

`c = (int)(c / ++c)`

Use int a=2, b=4, c=5, d=3;

```
5. (b = 2 * b) + (b + b)
```

Use int a=2, b=4, c=5, d=3;

```
6. int f = a>0 ? (int)2.7 : (int)3.7;
```

Use int a=2, b=4, c=5, d=3;

```
7. int[] g = {6,4,2,9,3,2,0,6,8,9,5};
   out.println( g[ g[b] + g[a] ] );
```

Use int a=2, b=4, c=5, d=3;

```
8. int[] g = {6,4,2,9,3,2,0,6,8,9,5};
   int[] h = {23, 49, 1, 14, 93};
   out.println( g[ g[b] + (g=h)[a] ] );
```